

Calculus with Parametric and Polar Curves



Parametric derivatives and arc length

- 1 For the curve $x = t^2$, $y = t^3$, find $\frac{dy}{dx}$ in terms of t .
 - 2 Find the arc length of the curve $x = 3t^2$, $y = 2t^3$ for $0 \leq t \leq 2$.
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Polar area

- 3 Find the area enclosed by one loop of the rose $r = 3\sin 2\theta$ (for $0 \leq \theta \leq \pi/2$).
 - 4 Find the area enclosed by the cardioid $r = 2 + 2\cos \theta$.
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Second derivatives and concavity for parametric curves

- 5 For $x = t^2$, $y = t^3 - 3t$, find $\frac{d^2y}{dx^2}$ in terms of t , using $\frac{d^2y}{dx^2} = \left[\frac{d}{dt} \left(\frac{dy}{dx} \right) \right] \div \frac{dx}{dt}$.
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Area between polar curves

- 6 Find the area inside the circle $r = 3$ and outside the circle $r = 2$ (an annular region).
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Horizontal and vertical tangents on parametric curves

- 7 For $x = t^2 - 4$, $y = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal ($\frac{dy}{dt} = 0$, $\frac{dx}{dt} \neq 0$).
 - 8 For the same curve, find the value(s) of t where the tangent line is vertical ($\frac{dx}{dt} = 0$, $\frac{dy}{dt} \neq 0$).
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Converting between polar and Cartesian for calculus

- 9 Find the slope of the tangent line to $r = 1 + \sin \theta$ at $\theta = 0$, using $x = r\cos \theta$, $y = r\sin \theta$ as parametric equations in θ .
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Speed and length for parametric motion

- 10 A particle moves according to $x(t) = t^2$, $y(t) = 4t$. Find its speed at $t = 3$.
- 11 Find the arc length of $x = \cos^3 t$, $y = \sin^3 t$ for $0 \leq t \leq \frac{\pi}{2}$ (an astroid arc), using $\frac{dx}{dt} = -3\cos^2 t \sin t$ and $\frac{dy}{dt} = 3\sin^2 t \cos t$.